

Test Report: Water Distribution System Performance

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Project: Substrate Mixer and Portioning Device

Component: Orifices (nozzles) of manifold

Prototype Version: V1 (5 holes), V2 (11 holes)

1 PART I — TEST PLAN

1.1 Research Hypotheses

The system consists of a manifold distributor ($L = 0.8$ m, $d \approx 3/4''$) with discrete discharge orifices fed by hydrostatic pressure. The primary objective is to achieve a uniform volumetric flow rate distribution (Q_i) across all orifices i .

In such systems, two competing physical phenomena influence the pressure gradient $p(x)$ along the pipe:

1. **Frictional Pressure Drop:** Viscous losses along the pipe wall reduce the available head.
2. **Mass Extraction:** The continuous removal of fluid through nozzles reduces the flow velocity in the main pipe, which can lead to a partial pressure recovery (Bernoulli effect).

The goal is to identify a configuration where the flow variance remains within $\pm 10\%$. We hypothesize that the exit velocity follows Bernoulli's principle. For an inviscid, incompressible, and steady flow, the conservation of energy is expressed as (Kundu et al. 2016, p. 299):

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \quad (1)$$

From this, a scaling law can be derived to relate the nozzle diameter d to the hydraulic head Δh :

$$d = C \cdot \Delta h^{-1/4} \quad (2)$$

where C is a system-specific constant.

1.2 Test Objectives

- Evaluate the impact of nozzle quantity on pressure drop and flow uniformity.
- Measure the volumetric flow rate at varying hydraulic heads ($\Delta h = 15$ cm to 120 cm).
- Determine the empirical constant C for the system's scaling law.

1.3 Measurement Methodology

The distribution pipe was mounted at a fixed height. A reservoir provided a constant head Δh . Measuring cylinders were placed under each nozzle. Water was allowed to flow for a certain time interval, after which the volume in each cylinder was recorded. For the 11-hole tests, due to spacing constraints, adjacent nozzles were grouped into single measurement vessels and the average per hole was calculated.

2 PART II — TEST EXECUTION DATA

2.1 Test 1: Baseline

Conditions: $\Delta h = 120$ cm, $t = 2$ min, $n_{\text{holes}} = 5$, $d = 1$ mm.

Hole ID	Volume [ml]
1	25
2	140
3	150
4	140
5	150

Table 1: Flow volumes for Test 1.

Calculated Parameters:

- Total flow volume: 605 ml (Avg: 121 ml/hole)
- Total flow rate: $Q = 302.5$ ml/min (5.04×10^{-6} m³/s)
- Exit velocity (avg): $v = 1.28$ m/s
- Max Relative Deviation: 83.3% (Signifies a blockage in Hole 1).

2.2 Test 2: Baseline – Maintenance Validation

Conditions: $\Delta h = 120$ cm, $t = 2$ min, $n_{\text{holes}} = 5$, $d = 1$ mm, Hole 1 cleaned.

Hole ID	Trial 1 [ml]	Trial 2 [ml]
1	200	225
2	180	190
3	170	185
4	170	190
5	160	200

Table 2: Flow volumes for Test 2.

Calculated Parameters:

- Flow rate Q : 440.0 – 495.0 ml/min
- Exit velocity v : 1.87 – 2.10 m/s
- Max Relative Deviation: 20.0% (Trial 1), 17.8% (Trial 2).

2.3 Test 3: Doubled Orifices

Conditions: $\Delta h = 15$ cm, $t = 2$ min, $n_{\text{holes}} = 11$, $d = 1$ mm.

Note: For Trial 2, the hose connection to the water reservoir was improved and sealed.

Group	Volume 1/2 [ml]	Hole ID	Processed 1/2 [ml]
1–2	130 / 175	1, 2	65.0 / 87.5
3–5	200 / 240	3, 4, 5	66.7 / 80.0
6–7	145 / 175	6, 7	72.5 / 87.5
8–9	150 / 175	8, 9	75.0 / 87.5
10–11	145 / 140	10, 11	72.5 / 70.0

Table 3: Raw and processed data for Test 3.

Calculated Parameters:

- Flow rate Q : 384.5 – 457.0 ml/min
- Exit velocity v : 0.74 – 0.88 m/s
- Max Relative Deviation: 13.3% (Trial 1), 20.0% (Trial 2).

2.4 Test 4: Doubled Orifices – Hydraulic Head Optimized

Conditions: $\Delta h = 30$ cm, $t = 2$ min, $n_{\text{holes}} = 11$, $d = 1$ mm.

Group	Volume [ml]	Hole ID	Processed [ml]
1–2	240	1, 2	120.0
3–5	440	3, 4, 5	146.7
6–7	245	6, 7	122.5
8–9	250	8, 9	125.0
10–11	220	10, 11	110.0

Table 4: Raw and processed data for Test 4.

Calculated Parameters:

- Total flow volume: 1395 ml
- Total flow rate Q : 697.5 ml/min (1.16×10^{-5} m³/s)
- Exit velocity v : 1.35 m/s
- Max Relative Deviation: 28.9%.

2.5 Empirical Scaling Law Analysis

To map out a predictable fluid distribution design, both an analytical loss model and experimental performance metrics were evaluated.

1. **Theoretical Approach (Discrete Attenuation Model):** To fulfill processing specifications, the system must distribute a target flow rate of $Q_{\text{inlet}} = 4.0$ L/min (6.667×10^{-5} m³/s) evenly across $N = 11$ discrete nozzles under a design hydraulic head of $\Delta h = 0.4$ m. Based on the steady, incompressible Bernoulli equation with discrete mass extractions (Kundu et al. 2016, p. 299), the exit velocity at the first nozzle is established as $v_1 = \sqrt{2g\Delta h}$.

To maintain the target substrate moisture content within the strict biochemical range of 60–70 wt%, the total spatial flow variance from the front to the rear of the mixer cannot

exceed 10%. This condition is mathematically satisfied by applying a discrete velocity attenuation factor $r = 0.99$ at each successive nozzle index, such that the discrete velocity profile expands as:

$$v_n = v_1 \cdot r^{n-1} \quad \text{where } r = 0.99, n \in [1, 11] \quad (3)$$

The terminal nozzle velocity is thus $v_{11} = v_1 \cdot (0.99)^{10} \approx 0.904v_1$, conforming to the 10% variance limit. Utilizing the conservation of mass, the cumulative volumetric flow rate is the sum of the discrete discharge streams:

$$Q_{\text{tot}} = A_{\text{orifice}} \sum_{n=1}^N v_n = \frac{\pi d_{\text{th}}^2}{4} \sum_{n=1}^{11} \left(\sqrt{2g\Delta h} \cdot 0.99^{n-1} \right) \quad (4)$$

Rearranging the state equation to solve directly for the ideal theoretical bore diameter d_{th} yields:

$$d_{\text{th}} = \left(\frac{4Q_{\text{tot}}}{\pi \sum_{n=1}^{11} \sqrt{2g\Delta h} \cdot 0.99^{n-1}} \right)^{1/2} \quad (5)$$

Substituting $g = 9.81 \text{ m/s}^2$, $\Delta h = 0.4 \text{ m}$, and $Q_{\text{tot}} = 6.667 \times 10^{-5} \text{ m}^3/\text{s}$, the summation of the velocity terms yields $\sum v_n \approx 29.32 \text{ m/s}$. This results in:

$$d_{\text{th}} = \left(\frac{4 \cdot (6.667 \times 10^{-5})}{\pi \cdot 29.32} \right)^{1/2} \approx 0.00170 \text{ m} \quad (6)$$

This model determines an ideal theoretical orifice diameter of **1.70 mm** to fulfill the customer throughput requirements while adhering to the distribution uniformity constraints.

2. **Empirical Approach:** Using the physical prototype diameter ($d = 1 \text{ mm}$), the calculated constants for the valid test runs showed extreme variance, ranging from 0.00062 to 0.00104. The derived empirical values are considered statistically unreliable for a rigorous scaling law. The high fluctuation in data suggests that at a 1 mm diameter, the flow is not governed solely by the macroscopic hydraulic head, but is dominated by stochastic factors such as nozzle-specific friction and manufacturing artifacts. Consequently, a deterministic scaling law cannot be established from the current dataset.

3 PART III — TEST REVIEW

3.1 Observations: Tests 1–2 (Initial Baseline)

The 5-hole configuration at $\Delta h = 120 \text{ cm}$ demonstrated sufficient total flow. However, Test 1 revealed extreme sensitivity to manufacturing artifacts. Residual burrs from the drilling process and internal pipe debris caused immediate partial blockages. This highlights that for small-diameter orifices ($d \leq 1 \text{ mm}$), rigorous deburring and system flushing are critical to prevent non-linear flow resistance.

3.1.1 System Upscaling after Test 2

Neither the empirical data (due to high variance) nor visual inspection indicated a significant pressure gradient along the length of the manifold. No consistent decrease in flow rate was observed from the inlet to the distal end. Based on the assumption that the internal pipe pressure drop remains negligible relative to nozzle resistance, the number of orifices was increased from 5 to 11 to evaluate the system's capacity limit.

3.2 Observations: Tests 2–4 (High-Density Configuration)

The transition to 11 holes (doubling the total discharge area) confirmed several key insights:

- **Stochastic Flow Distribution:** The flow variance between the 11 nozzles appeared random rather than following a linear pressure-drop gradient. This suggests that the 1 mm diameter is too small, making the orifices susceptible to partial clogging by microscopic burrs or fluid impurities.
- **Hydraulic Capacity:** Visual assessment showed no degradation in flow quality when moving from 5 to 11 holes, even at a low head of 15 cm. This indicates that the manifold’s cross-sectional area is sufficient to feed 11 nozzles without significant internal head loss.
- **Economic Efficiency:** Since the current configuration (11 holes, 1 mm) is far from the hydraulic limit, there is significant potential for diameter enlargement. In an industrial context, increasing the diameter to 2 mm would significantly enhance volumetric throughput, reducing mixing time and labor costs.

3.3 Interpretation & Uncertainty

Measurement uncertainty was primarily introduced by the grouping of nozzles into shared collection vessels during the 11-hole tests. While this facilitated data collection, it likely masked individual nozzle fluctuations. However, the qualitative and quantitative evidence suggests that flow performance is primarily head-dependent and that the current orifice size is the primary bottleneck.

3.4 Final Recommendations

Based on the geometric constraints of the assembly (170 cm max height vs. 130 cm mixer height), a maximum hydraulic head of $\Delta h = 40$ cm is available for the operating cycle. To maximize volumetric flow and ensure a more stable pressure gradient, the reservoir should be positioned at the maximum elevation of 170 cm.

To eliminate the systemic errors identified during testing, the following modification roadmap is established:

- **Stepwise Diameter Enlargement:** The manifold orifices must be expanded in systematic increments from 1.0 mm to 2.0 mm, 2.5 mm, and ultimately 3.0 mm. Moving past the 2.0 mm threshold is critical to ensure that micro-burrs and fluid particulates no longer dictate local discharge coefficients.
- **High-Fidelity Discrete Metering:** Subsequent validation cycles must abandon collective vessel grouping. Each of the 11 orifices must be metered via dedicated, isolated graduated cylinders to capture true individual spatial deviations along the manifold.
- **Process Optimization:** Implementing a final 3.0 mm bore layout will drastically elevate volumetric throughput. Shortening the moisture injection cycle directly minimizes manual labor times, compounding the operational efficiency and commercial viability of the device. However, since this significantly exceeds the hydraulic limit, a detailed metric-based analysis of this configuration must be performed.

3.5 Hypothesis Evaluation

The initial research hypothesis positing that the fluid discharge follows a highly predictable macro-hydraulic scaling law ($d = C \cdot \Delta h^{-1/4}$) is rejected for the sub-2.0 mm regime.

At a 1.0 mm diameter, the fluid dynamics are heavily decoupled from deterministic hydrostatic head equations. Instead, the flow distribution is strictly limited by micro-scale manufacturing quality. Variations in drilling tolerances, micro-burrs along the internal orifice boundaries, and trapped dust particles impose localized, stochastic boundary layers that outweigh potential flow benefits. True predictable fluid performance can only be recovered by enlarging the openings to scale past these mechanical surface defects.

References

Kundu, Pijush K. et al. (2016). *Fluid Mechanics*. Sixth. Elsevier Inc. ISBN: 978-0-12-405935-1.